Chapter 4

Using GenAI in the Real World

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As glitzy and even glamorous as GenAI and its outputs can be, it's the usefulness of this tool in managing everyday life that tends to prove its value. Or not. It's the changes to the routine and the mundane — the things you tend to view as constants in your personal and work life — that will be the most disrupted by GenAI and the most challenging to navigate. Life, as you know it, will change in a multitude of ways. Some will be profound, some will be barely noticeable, and others will be harmful.

In this chapter, you gain some insight into what some of these changes are likely to be. There will be more. Many more. But for now, consider these few earth-shaking impacts.

Dying Keywords

ChatGPT and its competitors, such as Microsoft Copilot, Google Gemini, and Meta AI, are already reshaping how people interact with search engines and use search engine optimization (SEO) to make content and websites searchable.

Because ChatGPT and the others can provide direct, summarized, and unified answers, many will accept those outputs as answers to their search query and look no further. In other words, they'll frequently bypass traditional search engines entirely. And when they do visit a search engine, they'll often be disinclined to click on traditional search results to investigate the matter further — or to fact-check the AI's response.

This hard shift in user behavior challenges the business model of search engines, which have long relied on users clicking on links. If too few people regularly click on links in search engine results, website traffic could diminish, which would also negatively affect ad-based revenue models.

Diminishing website traffic fallout

For example, Google may suffer a drop in traditional search ad revenue if people don't click on ads in the search results or click on ads posted on partner websites. Google faces a dilemma. To summarize, Google may be impacted negatively either way. Google Search can't compete with the likes of Bing Copilot and Perplexity AI search with its instant citation of sources for its AI responses if Google doesn't provide an AI alternative. So, of course, it does provide AI summaries at the top of some search results. But in doing so, it could be hurting its own revenue streams.

ChatGPT and competitor summaries at the head of search results bring no joy to any person or organization seeking to attract online traffic to their content or website. Web traffic may slow or even dry up in the future. The SEO keywords that these people long relied upon to bring traffic will lose their appeal to searchers because generative AI chatbots don't need keywords to understand and answer your question.

The one saving grace is that AI will continue to use keywords to find and pull information off the internet and out of private databases. ChatGPT

and its competitors will use other techniques, too, but the point is that search capabilities are still necessary at least for the moment.

Evolving SEO tactics

Meanwhile, SEO tactics will evolve as ranking in traditional search results becomes less critical, but machines will still use them. Content creators will need to focus on ensuring their material is easily accessible to AI applications and systems. Organized, machine-readable data (structured data), information enriched with clear context and meaning (semantic content), and indicators of credibility and trustworthiness (authoritative signals) will become increasingly important as AI tools like ChatGPT aggregate insights from a variety of sources."

This shift in user behavior on traditional search engines is likely to change how content is presented online as well. For example, consider a marketing content funnel, which is a common marketing strategy that uses targeted content to guide potential buyers from general awareness of a subject to the purchase of a specific product. This tunnel has three parts:

- **The top of the funnel** consists of general interest content designed to attract potential customers early in the buying cycle and links to content further down the funnel.
- **The middle of the funnel** offers content that contains more in-depth information and nurtures sales leads.
- **The bottom of the funnel** is filled with content focused on completing the sale of a specific product or service.



REMEMBER Top-of-the-funnel content will likely disappear as ChatGPT's narrative summaries can easily and quickly answer any requests for information on almost any topic. This has the potential of diminishing the draw of the tunnel strategy since search narratives will not point to or link to marketing content in any company's funnel.

Over-relying on AI-curated information

ChatGPT and other AI chatbots' ability to curate information may also reduce the time that you and others spend browsing individual websites. This might severely disrupt content consumption patterns and challenge businesses reliant on search-driven traffic.

Unfortunately, over-reliance on search summaries will likely add to the rise of misinformation, too. When people blindly accept AI summaries and outputs as factual, they're easily fooled. It's important to fact-check and verify generative AI outputs — ChatGPT results or otherwise — but that's not likely to happen on an appreciable scale because people tend to opt for convenience over substance.

As generative AI tools like ChatGPT continue to disrupt search results and content delivery, traditional SEO and monetization strategies will need to adapt. Bottom line: This fundamental shift in search behavior can mean that not only are keywords dying in the public's eye, but so may various types of online businesses and industries.

Moving from Information Search to Knowledge Assistants

As I note in the previous section, ChatGPT and similar AI models are driving a shift from traditional information search to the use of AI summaries as a unified search result. In so doing, these AI chatbots are also driving a swifter evolution in knowledge assistants. You're most likely familiar with earlier forms of knowledge assistants, such as Siri, Alexa, and Google Assistant. Instead of searching through links, you can ask questions and receive direct and unified answers, and that effectively transforms search into a conversation.

Now ChatGPT and similar AI chatbots are being embedded in these same knowledge assistants to supercharge their evolution. Thus, each can then provide more personalized, interactive, and contextualized responses.

Finding new purposes for knowledge assistants

But knowledge assistants are also expanding into new areas with new purposes. They can be built to help customers and employees or even serve the elite — from celebrities and CEOs to specialists in medicine, science, and space exploration. Each of these knowledge assistants has access to specialized and personalized data.

In general, knowledge assistants designed for highly specialized and customized service share a main purpose: delivering instant answers from a company's internal data sources through a conversational interface. They can comprehend queries and sift through knowledge bases, databases, and other internal repositories to provide precise answers and detailed processes. In doing so, both the work quality and the speed are greatly enhanced, which assists you in making informed decisions without having to constantly stop to manually search for information or reach out to colleagues.



WARNING Be careful not to confuse terms while you're sorting the various generative AI tools and functions. For example, agent assist tools suggest specific actions to facilitate work done by human helpdesk and customer service agents, but knowledge assistants are AI applications that focus on supplying information — either to the human agent working at the helpdesk or in IT support or directly to the customer. Knowledge assistants facilitate access to data, while agent assist tools manage the workflow. Another difference: Knowledge assistants allow for follow-up questions, suggest related queries, offer analytics on trends and effectiveness, and identify knowledge gaps. AI agent assist tools just focus on performing specific tasks.

Noting the pros and cons of knowledge assistants

The pros of the shift to knowledge assistants include faster access to information, convenience, and the ability to get detailed, contextual responses without sifting through multiple sources. Knowledge assistants can also handle more complex queries than traditional AI assistants by

combining information from different sources and from a specialized perspective.

However, cons come into this shift, too. People may become overly reliant on or even addicted to AI systems and tools like ChatGPT-enabled knowledge assistants. That could severely reduce critical thinking and independent research skills and activities. Transparency, accuracy, and bias are ongoing concerns with AI systems and tools; they may not always provide the most reliable or diverse sources of information. Additionally, the lack of direct interaction with websites may overly limit users' exposure to diverse perspectives and in-depth knowledge. In short, users may be dumbed down.

Living with Misinformation and Manipulation

Society has entered a new era of widespread false and misleading information. You need only look at the multitude of conspiracy theories zipping across the web on any given day to see that. But with GenAI, information dissemination jumps to an entirely new level, with significant implications in the spread of disinformation.

People can exploit generative AI chatbots to create convincing — but entirely fabricated — news articles, social media posts, and fake testimonials that are indistinguishable from legitimate content online. Consider these examples:

- **Anyone can prompt ChatGPT to fabricate a news story about a political figure involved in a scandal that never occurred.** The story, replete with fabricated quotes and false evidence, can spread rapidly across social media platforms, sowing confusion and manipulating public perception. It could even tilt an election, especially if many such falsely generated stories are spewed by these chatbots in every language and in every internet nook and cranny where targeted audiences can be reached.
- **The underlying generative AI models can be fine-tuned to mimic the writing style of specific individuals or reputable news outlets,** making the distinction between authentic and fake content increasingly difficult for the average person. Malicious actors can, for example, manipulate stock prices, influence election outcomes, and incite social unrest through a series of tweets and

other social media posts and comments that appear to be from a trusted journalist, a respected business leader, or a political figure.



WARNING The speed and scale at which GenAI chatbots can produce AI-generated disinformation poses significant challenges for social media platforms and individuals to overcome in the effort to maintain integrity in public discourse.

Manipulation tactics go beyond spreading false narratives to include strategies that create and maintain emotional and psychological impact. GenAI tools can analyze vast amounts of data to instantly identify the most effective ways to attract, engage, and manipulate specific individuals or groups. For example, GenAI tools can create polarizing or provocative material that's more likely to be shared and believed, even if it's entirely baseless. This capability can amplify existing divisions and create new ones within any given community. It can also undermine trust in news media, respectable sources, and the larger information ecosystem.



REMEMBER The line between reality and lies becomes increasingly blurred as GenAI tools continue to progress. The potential for misuse of these technologies requires the development and use of strong and continued countermeasures. Such countermeasures can include digital literacy education, the development of sophisticated detection tools, and the implementation of strict ethical guidelines governing the use of AI in the generation and dissemination of content.

Narrowing Options

If you're relying solely on ChatGPT or similar AI tools, be aware that it can limit your exposure to diverse perspectives, narrow your understanding of an issue or topic, and limit your awareness of differing evidence, opinions, and broader sources of information. Always remember that and make a point to look for other sources of information outside of

AI, or even through interactions with other AI tools that use a different underlying model than the first AI tool you used.



REMEMBER GenAI tools offer speed and convenience, but they do so at the cost of limiting your understanding of complex issues and your ability to connect the dots in information that can lead to new breakthroughs.

Recognizing the limitations of GenAI info

Lots of dangers for you and society can be found in overreliance on GenAI tools. Not all of them are obvious. For example, algorithmic bias is often invisible to your eye and your mind; nonetheless, it might cause you to unknowingly receive a skewed or incomplete picture of any issue or situation.

Further, when AI condenses complex topics into easily digestible AI summaries, you're losing out on the depth and nuance that may deliver an entirely different meaning than you gleaned from the AI's summary. It's as if you're trying to understand quantum physics but listening only to sound bites. For example, if you want to understand geopolitical conflicts, you may think you've got a good handle on the issues when you only have a simplified AI overview that excludes the historical context or conflicting ideologies involved. Over time, continued shallow exposures to information lead to a more surface-level or childlike understanding of important and complex subjects.

Looking only for confirmation

Confirmation bias, the tendency to interpret new evidence as confirmation of your own beliefs or theories, can also be a big problem. ChatGPT and similar generative AI tools generate responses based on patterns in your interactions with them. This can result in them delivering responses that inadvertently confirm and strengthen any preexisting viewpoints you may have.

If you consistently ask for information that aligns with your own biases, GenAI models will continuously oblige. You can even write prompts that inadvertently expose your belief or opinion in some way, which can cause a model to respond in the same vein, creating an echo chamber that seemingly confirms your stance. For example, if you aren't careful in your prompting while seeking political information, you may receive responses that mirror your views and ignore opposing arguments.

Many GenAI tools also don't routinely provide links to the original sources they use to generate a response. You can often prompt the model to provide that information for you — but even then, the list may be incorrect or even fabricated.

Exploiting varied information sources

Finally, using AI as the sole source of information limits the discovery of new information. You're more likely to find yourself covering familiar information repeatedly than you are to stumble upon new, unexpected ideas through exploring different sources. Traditional search often leads users to diverse content they weren't initially looking for, which can aid critical thinking and spur new ideas. By contrast, AI-generated responses tend to be more targeted and specific, which often eliminates any opportunity to find something new.

You can overcome this narrow perspective somewhat by prompting a GenAI tool to deliver related or associated information. That's still typically not as useful as free-thinking your way through search results, although those can limit your thinking, too. Every technological choice results in a trade-off of some sort. The more you allow technology to think for you or to assist you by hand-feeding you information, the more you limit your ability to expand your own knowledge.

Your Brain on GenAI

For better or worse, every technology affects the human brain in some way. For example, a study by McGill University in 2020 found that frequently using GPS causes a corresponding decline in hippocampal-depen-

dent spatial memory. In essence, you don't remember the way to places, so you rely on GPS to guide you again to the same place, and then you're stuck on repeat, each time becoming more reliant on GPS to get anywhere. Similarly, reliance on autocorrect diminishes your spelling skills, overuse of calculators softens math skills, and relying on your smartphone's contact database makes it hard for you to recite anyone's phone number.

Some technologies affect the human body as well, such as the effect too much screen viewing has on the eyes and the effect that always hunching over a smartphone or other device has on posture and neck pain.

Effects of GenAI on thinking skills

It should come as no surprise then that generative AI tools also affect you and society. For example, a research paper published in *Nature* titled "Impact of Artificial Intelligence on Human Loss in Decision Making, Laziness and Safety in Education" found that AI tools rob people of decision-making skills and lead to laziness. I would add that ChatGPT and other GenAI tools reduce intellectual curiosity and critical thinking skills. What's there to be curious about or contemplate when you know that anything you might want to know at any time is there for the asking? In other words, what is the purpose of extended knowledge exploration or pursuit if ChatGPT already has all the answers for you?

One of the more alarming aspects is that GenAI is supposed to free more of your time so you can spend it on higher purposes — presumably inventing, innovating, and creating stuff. But can you, though, if using these tools also reduces your cognitive ability?

In any case, it's good to be aware of what can happen so you can temper your use of this tech or take steps to offset the impact.



REMEMBER Cognitive atrophy is the elephant in the worry-room. Overreliance on GenAI tools to perform tasks such as writing, problem-solving, and even engaging in conversation might squash your ability to

perform these tasks without assistance. That's similar to, but more dire I think, than the "Google effect," aka "digital amnesia," that the Columbia University researchers detailed in the study "Google Effects on Memory: Cognitive Consequences of Having Information at Our Fingertips." Essentially, they found that the ease of finding information online impairs your ability to remember it. In the case of GenAI tools, you could soon find yourself struggling to carry on a conversation or write an email.

Effects of AI tools on interpersonal skills

Other researchers have found that overuse of conversational AI tools can affect social behavior and interpersonal skills. Essentially, you become more comfortable chatting with the AI tool than with another human. Sherry Turkle's research at MIT went even further by suggesting that people become isolated and lose their sense of empathy when they overuse technology.

But it's also important to realize that this discussion is about a tool and not a drug or a demon. You can control how and when you use it and balance your life so that you have plenty of interactions that don't involve AI or a device. And, as I mentioned before, GenAI tools bring many benefits. I've detailed many of them in other chapters of this book.

Protecting yourself from GenAI fallout

The key is to avoid thinking of GenAI tools as smarter than you. They aren't. They're just outstandingly good at data retrieval and data repackaging. Your brain, my friend, does way more than that.



TIP Don't rely excessively on this tech, but do use it like you would any other data repository, be that a dictionary, encyclopedias, the internet, your car's operating manual, whatever. Keep GenAI in its place and yourself in your place and venture boldly forth to make your own mark on the world. And when you do, take full credit. Generative AI tools are

no more the creator of your work than a hammer is for building an architect's design.

If, however, you let GenAI take the lead — to make your decisions and do your work — then it (and not you) receives the credit, although you'll still get the blame. Don't believe me? Just wait until you copy ChatGPT's homework only to discover it was completely wrong or outright plagiarism. Trust me, the court will summon you, not ChatGPT, to account and pay for the deed.